



**EXAM PAPERS PRACTICE**

Boost your performance and confidence  
with these topic-based exam questions

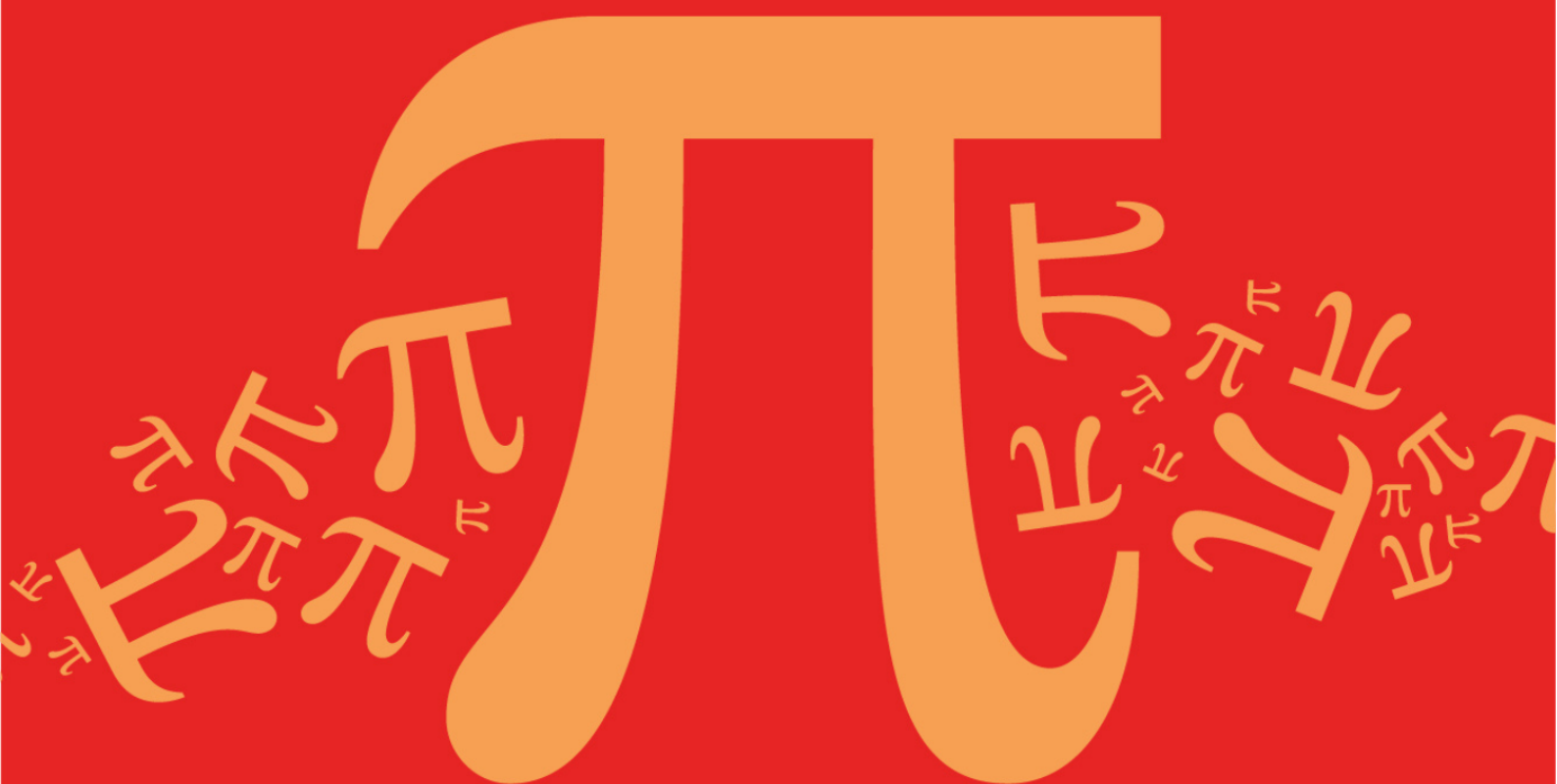
Practice questions created by actual  
examiners and assessment experts

Detailed mark scheme

Suitable for all boards

Designed to test your ability and  
thoroughly prepare you

# AQA A Level Further Mathematics (7367)



Topic

**C: Matrices**

Sub topic

**C1: Manipulate Matrices**

## Topic Questions

# AQA A Level Further Mathematics (7367)

## Question Paper

Topic

**C: Matrices**

Sub topic

**C1: Manipulate Matrices**

EXAM PAPERS PRACTICE

© 2024 Exams Papers Practice. All Rights Reserved

**To be used by all students and teachers for  
AQA A Level Further Mathematics (7367)  
for exams from 2025 onwards.**

Students and teachers of other boards may also find this useful

Name : \_\_\_\_\_

Date : \_\_\_\_\_

Time : \_\_\_\_\_

Total Marks Available : \_\_\_\_\_

Total Marks Received : \_\_\_\_\_

**Q1.**

Three matrices **A**, **B** and **C** are given by

$$\mathbf{A} = \begin{bmatrix} 5 & 2 & -3 \\ 0 & 7 & 6 \\ 4 & 1 & 0 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 1 & 0 \\ 3 & -5 \\ -2 & 6 \end{bmatrix} \quad \text{and} \quad \mathbf{C} = \begin{bmatrix} 6 & 4 & 3 \\ 1 & 2 & 0 \end{bmatrix}$$

Which of the following **cannot** be calculated?

Circle your answer.

**AB**

**AC**

**BC**

**A<sup>2</sup>**

(Total 1 mark)

**Q2.**

Two matrices **A** and **B** satisfy the equation

$$\mathbf{AB} = \mathbf{I} + 2\mathbf{A}$$

$$\mathbf{B} = \begin{bmatrix} 3 & -2 \\ -4 & 8 \end{bmatrix}$$

where **I** is the identity matrix and

Find **A**.

(Total 3 marks)

**Q3.**

Given that  $\mathbf{M} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$ , prove that  $\mathbf{M}^n = \begin{bmatrix} 3^{n-1} & 3^{n-1} & 3^{n-1} \\ 3^{n-1} & 3^{n-1} & 3^{n-1} \\ 3^{n-1} & 3^{n-1} & 3^{n-1} \end{bmatrix}$  for all  $n \in \mathbb{N}$

(Total 5 marks)

**Q4.**

(a) The matrix **X** is defined by  $\begin{bmatrix} 1 & 2 \\ 3 & 0 \end{bmatrix}$ .

(i) Given that  $\mathbf{X}^2 = \begin{bmatrix} m & 2 \\ 3 & 6 \end{bmatrix}$ , find the value of  $m$ .

(1)

(ii) Show that  $\mathbf{X}^3 - 7\mathbf{X} = n\mathbf{I}$ , where  $n$  is an integer and **I** is the  $2 \times 2$  identity matrix.

(4)

(b) It is given that  $\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ .

(i) Describe the geometrical transformation represented by **A**.

(1)

- (ii) The matrix **B** represents an anticlockwise rotation through  $45^\circ$  about the origin.

Show that  $\mathbf{B} = k \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$ , where  $k$  is a surd.

(2)

- (iii) Find the image of the point  $P(-1, 2)$  under an anticlockwise rotation through  $45^\circ$  about the origin, followed by the transformation represented by **A**.

(4)

(Total 12 marks)

### Q5.

The matrix **A** is given by

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & -1 \\ 1 & 2 & 1 \\ 2 & 2 & 3 \end{bmatrix}$$

- (a) Given that  $\mathbf{A}^2 = \begin{bmatrix} p & -2 & -4 \\ 5 & 6 & 4 \\ 10 & q & 9 \end{bmatrix}$ , find the value of  $p$  and the value of  $q$ .

(2)

- (b) Given that  $\mathbf{A}^3 - 6\mathbf{A}^2 + 11\mathbf{A} - 6\mathbf{I} = \mathbf{0}$ , prove that

$$\mathbf{A}^{-1} = \frac{1}{6}(\mathbf{A}^2 - 6\mathbf{A} + 11\mathbf{I})$$

(2)

©2025 Exam Papers Practice. All rights reserved.

- (c) Given that  $\mathbf{A}^{-1} = \frac{1}{6} \begin{bmatrix} r & -2 & 2 \\ -1 & 5 & -2 \\ -2 & s & 2 \end{bmatrix}$ , find the value of  $r$  and the value of  $s$ .

(2)

- (d) Hence, or otherwise, find the solution of the system of equations

$$\begin{aligned} x - z &= k \\ x + 2y + z &= 5 \\ 2x + 2y + 3z &= 7 \end{aligned}$$

giving your answers in terms of  $k$ .

(3)

(Total 9 marks)

### Q6.

The matrices **A** and **B** are defined by

$$\mathbf{A} = \begin{bmatrix} p & 2 \\ 4 & p \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} 3 & 1 \\ 2 & 3 \end{bmatrix}$$

(a) Find, in terms of  $p$ , the matrices:

(i)  $\mathbf{A} - \mathbf{B}$ ;

(1)

(ii)  $\mathbf{AB}$ .

(2)

(b) Show that there is a value of  $p$  for which  $\mathbf{A} - \mathbf{B} + \mathbf{AB} = k\mathbf{I}$ , where  $k$  is an integer and  $\mathbf{I}$  is the  $2 \times 2$  identity matrix, and state the corresponding value of  $k$ .

(4)

(Total 7 marks)

**Q7.**

The  $3 \times 3$  matrices  $\mathbf{A}$  and  $\mathbf{B}$  satisfy

$$\mathbf{AB} = \begin{bmatrix} k & 8 & 1 \\ 1 & 1 & 0 \\ 1 & 4 & 0 \end{bmatrix}, \text{ where } \mathbf{A} = \begin{bmatrix} k & 6 & 8 \\ 0 & 1 & 2 \\ -3 & 4 & 8 \end{bmatrix}$$

and  $k$  is a constant.

(a) Show that  $\mathbf{AB}$  is non-singular.

(1)

(b) Find  $(\mathbf{AB})^{-1}$  in terms of  $k$ .

(5)

(c) Find  $\mathbf{B}^{-1}$ .

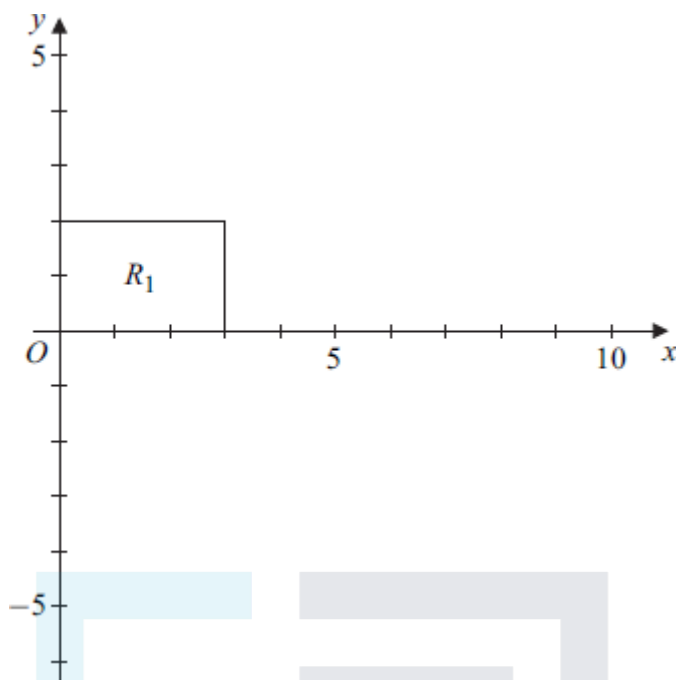
(4)

(Total 10 marks)

**Q8.**

The diagram below shows a rectangle  $R_1$  which has vertices  $(0, 0)$ ,  $(3, 0)$ ,  $(3, 2)$  and  $(0, 2)$ .

(a) On the diagram, draw:



- (i) the image  $R_2$  of  $R_1$  under a rotation through  $90^\circ$  clockwise about the origin; (1)

- (ii) the image  $R_3$  of  $R_2$  under the transformation which has matrix

$$\begin{bmatrix} 4 & 0 \\ 0 & 2 \end{bmatrix}$$

(3)

- (b) Find the matrix of:

- (i) the rotation which maps  $R_1$  onto  $R_2$ ; (1)

- (ii) the combined transformation which maps  $R_1$  onto  $R_3$ . (3)

(Total 8 marks)

**Q9.**

- (a) Using surd forms, find the matrix of a rotation about the origin through  $135^\circ$  anticlockwise. (2)

- (b) The matrix  $\mathbf{M}$  is defined by  $\mathbf{M} = \begin{bmatrix} -1 & -1 \\ 1 & -1 \end{bmatrix}$ .

- (i) Given that  $\mathbf{M}$  represents an enlargement followed by a rotation, find the scale factor of the enlargement and the angle of the rotation. (3)

- (ii) The matrix  $\mathbf{M}^2$  also represents an enlargement followed by a rotation. State the scale factor of the enlargement and the angle of the rotation.

- (2)
- (iii) Show that  $\mathbf{M}^4 = k\mathbf{I}$ , where  $k$  is an integer and  $\mathbf{I}$  is the  $2 \times 2$  identity matrix. (2)
- (iv) Deduce that  $\mathbf{M}^{2012} = -2^n \mathbf{I}$  for some positive integer  $n$ . (2)
- (Total 11 marks)

### Q10.

The matrix  $\mathbf{A} = \begin{bmatrix} k & 1 & 2 \\ 2 & k & 1 \\ 1 & 2 & k \end{bmatrix}$ , where  $k$  is a real constant.

- (a) (i) Show that there is a value of  $k$  for which
- $$\mathbf{A} \mathbf{A}^T = m \mathbf{I}$$
- where  $m$  is a rational number to be determined and  $\mathbf{I}$  is the  $3 \times 3$  identity matrix. (6)
- (ii) Deduce the inverse matrix,  $\mathbf{A}^{-1}$ , of  $\mathbf{A}$  for this value of  $k$ . (1)
- (b) (i) Find  $\det \mathbf{A}$  in terms of  $k$ . (2)
- (ii) In the case when  $\mathbf{A}$  is singular, find the integer value of  $k$  and show that there are no other possible real values of  $k$ . (3)
- (iii) Find the value of  $k$  for which  $\lambda = 7$  is a real eigenvalue of  $\mathbf{A}$ . (2)
- (Total 14 marks)

### Q11.

- (a) Write down the  $2 \times 2$  matrix corresponding to each of the following transformations:
- (i) a rotation about the origin through  $90^\circ$  clockwise; (1)
- (ii) a rotation about the origin through  $180^\circ$ . (1)
- (b) The matrices  $\mathbf{A}$  and  $\mathbf{B}$  are defined by

$$\mathbf{A} = \begin{bmatrix} 2 & 4 \\ -1 & -3 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} -2 & 1 \\ -4 & 3 \end{bmatrix}$$

- (i) Calculate the matrix  $\mathbf{AB}$ . (2)
- (ii) Show that  $(\mathbf{A} + \mathbf{B})^2 = k\mathbf{I}$ , where  $\mathbf{I}$  is the identity matrix, for some integer  $k$ . (3)
- (c) Describe the single geometrical transformation, or combination of two geometrical transformations, represented by each of the following matrices:
- (i)  $\mathbf{A} + \mathbf{B}$ ; (2)
- (ii)  $(\mathbf{A} + \mathbf{B})^2$ ; (2)
- (iii)  $(\mathbf{A} + \mathbf{B})^4$ . (2)
- (Total 13 marks)

## Q12.

$$\mathbf{X} = \begin{bmatrix} 3 & 1 & -3 \\ 2 & 4 & 3 \\ -4 & 2 & -1 \end{bmatrix}$$

The non-singular matrix

- (a) (i) Show that  $\mathbf{X}^2 - \mathbf{X} = k\mathbf{I}$  for some integer  $k$ . (3)
- (ii) Hence show that  $\mathbf{X}^{-1} = \frac{1}{20}(\mathbf{X} - \mathbf{I})$ . (2)
- ©2025 Exam Papers Practice. All rights reserved.

$$\mathbf{Y}^{-1} = \begin{bmatrix} 60 & 0 & 0 \\ 0 & 0 & -10 \\ 0 & 20 & 0 \end{bmatrix}$$

- (b) The  $3 \times 3$  matrix  $\mathbf{Y}$  has inverse

Without finding  $\mathbf{Y}$ , determine the matrix  $(\mathbf{XY})^{-1}$ .

(3)  
(Total 8 marks)

## Q13.

The matrix  $\mathbf{A}$  is defined by

$$\mathbf{A} = \begin{bmatrix} -1 & -\sqrt{3} \\ \sqrt{3} & -1 \end{bmatrix}$$

- (a) (i) Calculate the matrix  $\mathbf{A}^2$  (2)
- (ii) Show that  $\mathbf{A}^3 = k\mathbf{I}$ , where  $k$  is an integer and  $\mathbf{I}$  is the  $2 \times 2$  identity matrix.



(2)

- (b) Describe the single geometrical transformation, or combination of two geometrical transformations, corresponding to each of the matrices:

(i)  $\mathbf{A}^3$ ;

(2)

(ii)  $\mathbf{A}$ .

(3)

(Total 9 marks)

### Q14.

It is given that

$$\mathbf{A} = \begin{bmatrix} 1 & 4 \\ 3 & 1 \end{bmatrix}$$

and that  $\mathbf{I}$  is the  $2 \times 2$  identity matrix.

- (a) Show that  $(\mathbf{A} - \mathbf{I})^2 = k\mathbf{I}$  for some integer  $k$ .

(3)

- (b) Given further that

$$\mathbf{B} = \begin{bmatrix} 1 & 3 \\ p & 1 \end{bmatrix}$$

find the integer  $p$  such that

$$(\mathbf{A} - \mathbf{B})^2 = (\mathbf{A} - \mathbf{I})^2$$

(4)

(Total 7 marks)

### Q15.

The matrices  $\mathbf{A}$  and  $\mathbf{B}$  are defined in terms of a real parameter  $t$  by

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & 1 \\ 2 & t & 4 \\ 3 & 2 & -1 \end{bmatrix} \text{ and } \mathbf{B} = \begin{bmatrix} 15 & -4 & -1 \\ -2t & 4 & 2 \\ 17 & -4 & -3 \end{bmatrix}$$

- (a) Find, in terms of  $t$ , the matrix  $\mathbf{AB}$  and deduce that there exists a value of  $t$  such that  $\mathbf{AB}$  is a scalar multiple of the  $3 \times 3$  identity matrix  $\mathbf{I}$ .

(5)

- (b) For this value of  $t$ , deduce  $\mathbf{A}^{-1}$ .

(2)

(Total 7 marks)

**Q16.**

The plane transformations  $T_A$  and  $T_B$  are represented by the matrices **A** and **B**

respectively, where  $\mathbf{A} = \begin{bmatrix} 3 & -1 \\ -5 & 2 \end{bmatrix}$  and  $\mathbf{B} = \begin{bmatrix} 2 & 5 \\ 5 & 13 \end{bmatrix}$ .

- (a) Find the equation of the line which is the image of  $y = 2x + 1$  under  $T_A$ .

(3)

- (b) The rectangle  $PQRS$ , with area  $4.5 \text{ cm}^2$ , is mapped onto the parallelogram  $P'Q'R'S'$  under  $T_B$ . Determine the area of  $P'Q'R'S'$ .

(2)

- (c) The transformation  $T_C$  is the composition

' $T_B$  followed by  $T_A$ '

Find the matrix which represents  $T_C$ , give a full geometrical description of  $T_C$ .

(5)

(Total 10 marks)

**Q17.**

The matrices **A** and **B** are defined by

$$\mathbf{A} = \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ 1 & 1 \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ 1 & 1 \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}$$

Describe fully the geometrical transformation represented by each of the following matrices: ©2025 Exam Papers Practice. All rights reserved.

- (a) **A**;

(2)

- (b) **B**;

(2)

- (c)  $\mathbf{A}^2$ ;

(2)

- (d)  $\mathbf{B}^2$ ;

(2)

- (e) **AB**.

(3)

(Total 11 marks)

**Q18.**

Let  $\mathbf{A} = \begin{bmatrix} 1 & x \\ 2 & 3 \end{bmatrix}$ ,  $\mathbf{B} = \begin{bmatrix} 1 & -1 \\ 2 & 2 \end{bmatrix}$  and  $\mathbf{C} = \begin{bmatrix} 4 - 4x & 8 \\ 8x - 4 & 4 \end{bmatrix}$ .

- (a) Find  $\mathbf{AB}$  in terms of  $x$ .

(2)

- (b) Show that  $\mathbf{B}^T \mathbf{A}^T = \mathbf{C}$  for some value of  $x$ .

(5)

(Total 7 marks)

### Q19.

The matrices  $\mathbf{A}$  and  $\mathbf{B}$  are defined by

$$\mathbf{A} = \begin{bmatrix} k & k \\ k & -k \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} -k & k \\ k & k \end{bmatrix}$$

where  $k$  is a constant.

- (a) Find, in terms of  $k$ :

(i)  $\mathbf{A} + \mathbf{B}$ ;

(1)

(ii)  $\mathbf{A}^2$ .

(2)

- (b) Show that  $(\mathbf{A} + \mathbf{B})^2 = \mathbf{A}^2 + \mathbf{B}^2$ .

(4)

- (c) It is now given that  $k = 1$ .

- (i) Describe the geometrical transformation represented by the matrix  $\mathbf{A}^2$ .

(2)

- (ii) The matrix  $\mathbf{A}$  represents a combination of an enlargement and a reflection. Find the scale factor of the enlargement and the equation of the mirror line of the reflection.

(3)

(Total 12 marks)

### Q20.

The  $2 \times 2$  matrices  $\mathbf{A}$  and  $\mathbf{B}$  are such that

$$\mathbf{AB} = \begin{bmatrix} 9 & 1 \\ 7 & 13 \end{bmatrix} \quad \text{and} \quad \mathbf{BA} = \begin{bmatrix} 14 & 2 \\ 1 & 8 \end{bmatrix}$$

Without finding  $\mathbf{A}$  and  $\mathbf{B}$ :

- (a) find the value of  $\det \mathbf{B}$ , given that  $\det \mathbf{A} = 10$ ;

(3)

- (b) determine the  $2 \times 2$  matrices  $\mathbf{C}$  and  $\mathbf{D}$  given by

$$\mathbf{C} = (\mathbf{B}^T \mathbf{A}^T) \quad \text{and} \quad \mathbf{D} = (\mathbf{A}^T \mathbf{B}^T)^T$$

where  $\mathbf{M}^T$  denotes the transpose of matrix  $\mathbf{M}$ .

(3)

(Total 6 marks)

**Q21.**

(a) Using surd forms where appropriate, find the matrix which represents:

(i) a rotation about the origin through  $30^\circ$  anticlockwise;

(2)

(ii) a reflection in the line  $y = \frac{1}{\sqrt{3}}x$ .

(2)

(b) The matrix  $\mathbf{A}$ , where

$$\mathbf{A} = \begin{bmatrix} 1 & \sqrt{3} \\ \sqrt{3} & -1 \end{bmatrix}$$

represents a combination of an enlargement and a reflection. Find the scale factor of the enlargement and the equation of the mirror line of the reflection.

(2)

(c) The transformation represented by  $\mathbf{A}$  is followed by the transformation represented by  $\mathbf{B}$ , where

$$\mathbf{B} = \begin{bmatrix} \sqrt{3} & -1 \\ 1 & \sqrt{3} \end{bmatrix}$$

Find the matrix of the combined transformation and give a full geometrical description of this combined transformation.

(5)

(Total 11 marks)

**Q22.**

Let  $\mathbf{P} = \begin{bmatrix} 1 & 4 & 2 \\ -1 & 2 & 6 \end{bmatrix}$  and  $\mathbf{Q} = \begin{bmatrix} k & 1 \\ 2 & -1 \\ 3 & 1 \end{bmatrix}$ , where  $k$  is a constant.

(a) Determine the product matrix  $\mathbf{PQ}$ , giving its elements in terms of  $k$  where appropriate.

(3)

(b) Find the value of  $k$  for which  $\mathbf{PQ}$  is singular.

(2)

(Total 5 marks)

**Q23.**

The matrix **M** is defined by

$$\mathbf{M} = \begin{bmatrix} \sqrt{3} & 3 \\ 3 & -\sqrt{3} \end{bmatrix}$$

- (a) (i) Show that

$$\mathbf{M}^2 = p\mathbf{I}$$

where  $p$  is an integer and  $\mathbf{I}$  is the  $2 \times 2$  identity matrix.

(3)

- (ii) Show that the matrix **M** can be written in the form

$$q \begin{bmatrix} \cos 60^\circ & \sin 60^\circ \\ \sin 60^\circ & -\cos 60^\circ \end{bmatrix}$$

where  $q$  is a real number. Give the value of  $q$  in surd form.

(3)

- (b) The matrix **M** represents a combination of an enlargement and a reflection.

Find:

- (i) the scale factor of the enlargement;

(1)

- (ii) the equation of the mirror line of the reflection.

(1)

- (c) Describe fully the geometrical transformation represented by  $\mathbf{M}^4$ .

(2)

©2025 Exam Papers Practice. All rights reserved.

(Total 10 marks)

## Q24.

The non-singular matrix  $\mathbf{M} = \begin{bmatrix} 2 & -1 & 1 \\ 1 & 0 & 1 \\ 1 & -1 & 2 \end{bmatrix}$ .

- (a) (i) Show that

$$\mathbf{M}^2 + 2\mathbf{I} = k\mathbf{M}$$

for some integer  $k$  to be determined.

(3)

- (ii) By multiplying the equation in part (a)(i) by  $\mathbf{M}^{-1}$ , show that

$$\mathbf{M}^{-1} = a\mathbf{M} + b\mathbf{I}$$

for constants  $a$  and  $b$  to be found.

- (3)
- (b) (i) Determine the characteristic equation of  $\mathbf{M}$  and show that  $\mathbf{M}$  has a repeated eigenvalue, 1, and another eigenvalue, 2. (6)
- (ii) Give a full set of eigenvectors for each of these eigenvalues. (5)
- (iii) State the geometrical significance of each set of eigenvectors in relation to the transformation with matrix  $\mathbf{M}$ . (3)
- (Total 20 marks)

### Q25.

The matrices  $\mathbf{A}$  and  $\mathbf{B}$  are given by

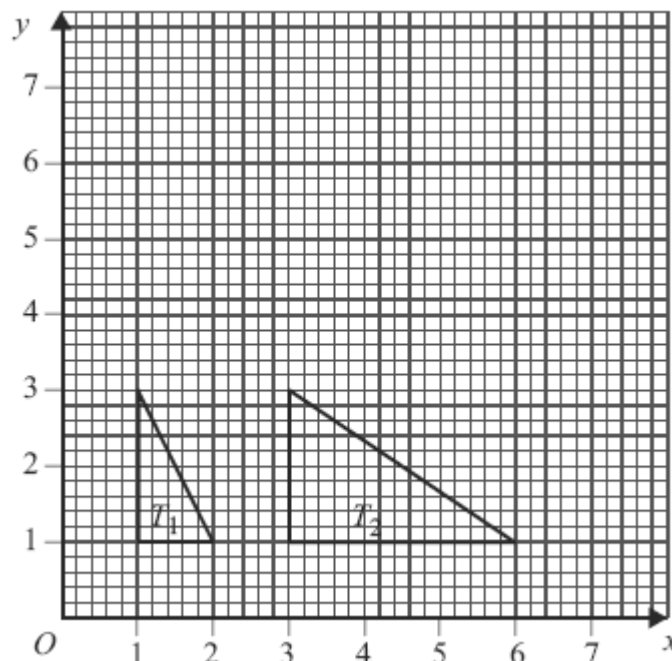
$$\mathbf{A} = \begin{bmatrix} 0 & 2 \\ 2 & 0 \end{bmatrix}, \mathbf{B} = \begin{bmatrix} 2 & 0 \\ 0 & -2 \end{bmatrix}$$

- (a) Calculate the matrix  $\mathbf{AB}$ . (2)
- (b) Show that  $\mathbf{A}^2$  is of the form  $k\mathbf{I}$ , where  $k$  is an integer and  $\mathbf{I}$  is the  $2 \times 2$  identity matrix. (2)
- (c) Show that  $(\mathbf{AB})^2 \neq \mathbf{A}^2\mathbf{B}^2$ . (3)
- (Total 7 marks)

©2025 Exam Papers Practice. All rights reserved.

### Q26.

The diagram shows two triangles,  $T_1$  and  $T_2$ .



- (a) Find the matrix of the stretch which maps  $T_1$  to  $T_2$ . (2)
- (b) The triangle  $T_2$  is reflected in the line  $y = x$  to give a third triangle,  $T_3$ .  
On the figure above, draw the triangle  $T_3$ . (2)
- (c) Find the matrix of the transformation which maps  $T_1$  to  $T_3$ . (3)
- (Total 7 marks)

### Q27.

The matrices **A** and **B** are given by

$$\mathbf{A} = \begin{bmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} \frac{\sqrt{3}}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{\sqrt{3}}{2} \end{bmatrix}$$

- (a) Calculate:
- (i)  $\mathbf{A} + \mathbf{B}$ ; (2)
- (ii)  $\mathbf{BA}$ . (3)
- (b) Describe fully the geometrical transformation represented by each of the following matrices:
- (i)  $\mathbf{A}$ ; (2)
- (ii)  $\mathbf{B}$ ; (2)
- (iii)  $\mathbf{BA}$ . (2)
- (Total 11 marks)

### Q28.

The matrices  $\mathbf{M}_A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix}$  and  $\mathbf{M}_B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$  represent the transformations A and B respectively.

- (a) Give a full geometrical description of each of A and B. (5)
- (b) Transformation C is obtained by carrying out A followed by B.

- (i) Find  $\mathbf{M}_C$ , the matrix of  $C$ . (2)
- (ii) Hence give a full geometrical description of the single transformation  $C$ . (2)
- (Total 9 marks)

**Q29.**

The matrix  $\mathbf{P} = \begin{bmatrix} 4 & -1 & 2 \\ 1 & 1 & 3 \\ -2 & 0 & a \end{bmatrix}$ , where  $a$  is constant.

- (a) (i) Determine  $\det \mathbf{P}$  as a linear expression in  $a$ . (2)
- (ii) Evaluate  $\det \mathbf{P}$  in the case when  $a = 3$ . (1)
- (iii) Find the value of  $a$  for which  $\mathbf{P}$  is singular. (2)
- (b) The  $3 \times 3$  matrix  $\mathbf{Q}$  is such that  $\mathbf{PQ} = 25\mathbf{I}$ .

Without finding  $\mathbf{Q}$ :

- (i) write down an expression for  $\mathbf{P}^{-1}$  in terms of  $\mathbf{Q}$ ; (1)
- (ii) find the value of the constant  $k$  such that  $(\mathbf{PQ})^{-1} = k\mathbf{I}$ ; (2)
- (iii) determine the numerical value of  $\det \mathbf{Q}$  in the case when  $a = 3$ . (4)

(Total 12 marks)

**Q30.**

The matrices  $\mathbf{A}$  and  $\mathbf{B}$  are given by

$$\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 3 & 8 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

The matrix  $\mathbf{M} = \mathbf{A} - 2\mathbf{B}$ .

- (a) Show that  $\mathbf{M} = n \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}$ , where  $n$  is a positive integer. (2)
- (b) The matrix  $\mathbf{M}$  represents a combination of an enlargement of scale factor  $p$  and a reflection in a line  $L$ . State the value of  $p$  and write down the equation of  $L$ .



(2)

- (c) Show that

$$\mathbf{M}^2 = q\mathbf{I}$$

where  $q$  is an integer and  $\mathbf{I}$  is the  $2 \times 2$  identity matrix.

(2)

(Total 6 marks)

### Q31.

The matrices  $\mathbf{A}$  and  $\mathbf{B}$  are given by

$$\mathbf{A} = \begin{bmatrix} 1 & 1 \\ -1 & 1 \\ 1 & 1 \end{bmatrix} \quad \text{and} \quad \mathbf{B} = \begin{bmatrix} 1 & 0 & 2 \\ 2 & 2 & t \end{bmatrix}$$

- (a) Find, in terms of  $t$ , the matrices:

(i)  $\mathbf{AB}$ ;

(3)

(ii)  $\mathbf{BA}$ .

(2)

- (b) Explain why  $\mathbf{AB}$  is singular for all values of  $t$ .

(1)

- (c) In the case when  $t = -2$ , show that the transformation with matrix  $\mathbf{BA}$  is the combination of an enlargement,  $E$ , and a second transformation,  $F$ . Find the scale factor of  $E$  and give a full geometrical description of  $F$ .

(6)

(Total 12 marks)

### Q32.

- (a) The transformation  $T$  is defined by the matrix  $\mathbf{A}$ , where

$$\mathbf{A} = \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}$$

- (i) Describe the transformation  $T$  geometrically.

(2)

- (ii) Calculate the matrix product  $\mathbf{A}^2$ .

(2)

- (iii) Explain briefly why the transformation  $T$  followed by  $T$  is the identity transformation.

(1)

- (b) The matrix  $\mathbf{B}$  is defined by

$$\mathbf{B} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

(i) Calculate  $\mathbf{B}^2 - \mathbf{A}^2$ .

(3)

(ii) Calculate  $(\mathbf{B} + \mathbf{A})(\mathbf{B} - \mathbf{A})$ .

(3)

(Total 11 marks)

### Q33.

The matrices  $\mathbf{P}$  and  $\mathbf{Q}$  are defined in terms of the constant  $k$  by

$$\mathbf{P} = \begin{bmatrix} 3 & 2 & 1 \\ 1 & -1 & k \\ 5 & 3 & 2 \end{bmatrix} \text{ and } \mathbf{Q} = \begin{bmatrix} 5 & 4 & 1 \\ 3 & k & -1 \\ 7 & 3 & 2 \end{bmatrix}$$

(a) Express  $\det \mathbf{P}$  and  $\det \mathbf{Q}$  in terms of  $k$ .

(3)

(b) Given that  $\det(\mathbf{PQ}) = 16$ , find the two possible values of  $k$ .

(4)

(Total 7 marks)

### Q34.

The matrix  $\mathbf{M}$  is defined by

$$\mathbf{M} = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$$

(a) Find the matrix:

(i)  $\mathbf{M}^2$ ;

(3)

(ii)  $\mathbf{M}^4$ .

(1)

(b) Describe fully the geometrical transformation represented by  $\mathbf{M}$ .

(2)

(c) Find the matrix  $\mathbf{M}^{2006}$ .

(3)

(Total 9 marks)

### Q35.

The matrices  $\mathbf{P}$  and  $\mathbf{Q}$  are given by



$$\mathbf{P} = \begin{bmatrix} 2 & 1 & 1 \\ 1 & t & -2 \\ 3 & 2 & 1 \end{bmatrix} \quad \text{and} \quad \mathbf{Q} = \begin{bmatrix} 1 & 1 & 1 \\ -7 & -1 & 5 \\ 11 & -1 & -7 \end{bmatrix}$$

where  $t$  is a real constant.

- (a) Find the value of  $t$  for which  $\mathbf{P}$  is singular.

(2)

- (b) (i) Determine the matrix  $\mathbf{R} = \mathbf{PQ}$ , giving its elements in terms of  $t$  where appropriate.

(3)

- (ii) Find the value of  $t$  for which  $\mathbf{R} = k\mathbf{1}$ , for some integer  $k$ .

(2)

- (iii) Hence find the matrix  $\mathbf{Q}^{-1}$ .

(1)

- (c) In the case when  $t = -3$ , describe the geometrical transformation with matrix  $\mathbf{R}$ .

(2)

(Total 10 marks)

### Q36.

- (a) The transformation  $T_1$  is defined by the matrix

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Describe this transformation geometrically.

(2)

- (b) The transformation  $T_2$  is an anticlockwise rotation about the origin through an angle of  $60^\circ$ .

Find the matrix of the transformation  $T_2$ . Use surds in your answer where appropriate.

(3)

- (c) Find the matrix of the transformation obtained by carrying out  $T_1$  followed by  $T_2$ .

(3)

(Total 8 marks)

### Q37.

The matrices  $\mathbf{A}$  and  $\mathbf{B}$  are defined by

$$\mathbf{A} = \begin{bmatrix} 3 & 4 \\ 4 & 3 \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} 0 & 2 \\ 2 & 0 \end{bmatrix}$$

(a) Calculate the matrices:

(i)  $\mathbf{A} + \mathbf{B}$ ;

(2)

(ii)  $\mathbf{AB}$ .

(2)

(b) Show that  $\mathbf{A} + \mathbf{B} - \mathbf{AB} = k\mathbf{I}$ , where  $k$  is an integer and  $\mathbf{I}$  is the  $2 \times 2$  identity matrix.

(2)

(Total 6 marks)

**Q38.**

(a) Describe the transformation given by each of the matrices

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \text{ and } \mathbf{B} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$$

(4)

(b) (i) Determine the matrix  $\mathbf{AB}$ .

(2)

(ii) Given that  $\mathbf{AB}$  represents a rotation, find the axis of rotation and the magnitude of the angle of rotation.

(2)

(Total 8 marks)

EXAM PAPERS PRACTICE

©2025 Exam Papers Practice. All rights reserved.